

BostonGene IML-2 Test Task

Painting-Style Classification — Full Project Report

A decision-oriented, mathematically grounded summary of the complete pipeline: EDA and stratified splitting, supervised classification, self-supervised pretraining, a feature-space variational autoencoder, clustering, gradient boosting, and explainable-AI analysis.

Item	Value
Task	8-class image classification by painting style
Dataset	1,422 images, 0 corrupted, imbalance ratio 5.6:1
Split	Stratified 70/15/15 — train 995 / val 213 / test 214
Image size	128 x 128 after PadToSquare
Best classifier (Task 1)	ResNet-18 transfer — macro-F1 0.732
Best embedding model (Task 2)	DINOv2 feature-space VAE — latent 128-d
Best overall (Task 3)	XGBoost on VAE embeddings — macro-F1 0.865
XAI	Grad-CAM, t-SNE/UMAP, permutation importance

3. Task 1 — Supervised Classifier (Notebook 02)

3.1 Loss and metrics — formal definition

The loss is class-weighted cross-entropy; ResNet-18 additionally uses label smoothing (0.05) to curb over-confidence on a small dataset:

$$\mathcal{L}_{CE} = - \sum_{c=1}^C w_c y_c \log \hat{p}_c$$

Because the classes are imbalanced, the primary metric is macro-F1 (every class weighted equally), with balanced accuracy as a secondary check:

$$F1_{macro} = \frac{1}{C} \sum_{c=1}^C \frac{2 P_c R_c}{P_c + R_c}$$

$$\text{BalancedAcc} = \frac{1}{C} \sum_{c=1}^C \text{Recall}_c$$

3.2 Results and convergence

Model	Input	Loss	Acc.	Macro-F1	Bal. Acc.
SimpleCNN (scratch)	128	Weighted CE	0.542	0.495	0.517
ResNet-18 (transfer)	128	Weighted CE + smoothing	0.757	0.732	0.763

ResNet-18 fits training quickly (train macro-F1 nears 1.0) while validation stabilises at 0.65-0.68, best at epoch 13; early stopping preserves that checkpoint. The train/val gap is the overfitting risk inherent to 995 images. Transfer learning improved test macro-F1 by +0.238 absolute over the scratch CNN — the single insight that motivates the rest of the project: pretrained representations are essential here.



Figure 2. Test confusion matrices (cell values = counts, colour = row-normalised recall). ResNet-18 produces a far cleaner diagonal; residual errors concentrate among the natural-scene styles (Impressionism / Naturalism / Rococo / Japonism) that genuinely share subject matter and palette.

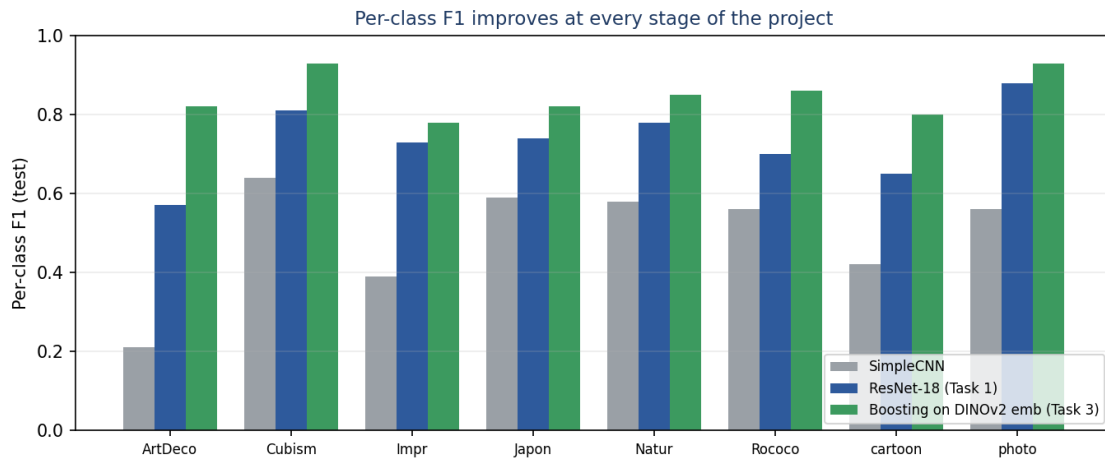


Figure 3. Per-class F1 at three points in the project. Every class improves monotonically from the scratch CNN to transfer learning to gradient boosting on DINOv2 embeddings; the largest gains are on the hardest minority class, ArtDeco (0.21 -> 0.57 -> 0.82).

4. Task 1 Bonus — Self-Supervised Learning (Notebook 02b)

A SimCLR encoder was pretrained without labels on the 995 training images only. It learns to pull together two augmentations of the same image and push apart all others in the batch, via the NT-Xent contrastive loss over cosine similarities:

$$\text{sim}(z_i, z_j) = \frac{z_i \cdot z_j}{\|z_i\| \|z_j\|}$$

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)}$$

NT-Xent: tau is the temperature; the denominator sums over all 2N augmented views in the batch, so larger batches give more negatives and a stronger signal.

Pretraining loss fell from 4.26 to 3.18 (objective optimised). The frozen encoder was then evaluated by linear probe and fine-tuning:

Evaluation	Accuracy	Macro-F1	Bal. Acc.
SimCLR + frozen linear probe	0.388	0.361	0.381
SimCLR + fine-tuning	0.481	0.468	0.489
(reference) ResNet-18 transfer	0.757	0.732	0.763

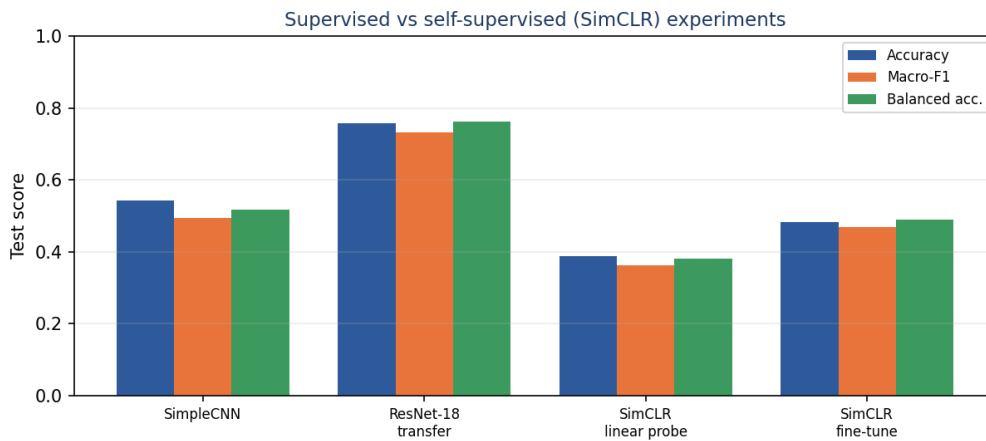


Figure 4. SimCLR underperforms ImageNet transfer learning — the expected result, since ImageNet uses 1.2M images whereas this contrastive model sees only 995 in-domain images, far below the data/batch scale contrastive methods need. Presented as a representation-learning demonstration, not the primary model.

5. Task 2 — Variational Autoencoder Embeddings (Notebooks 03 / 03b)

5.1 Mathematical formulation

A VAE maximises the Evidence Lower Bound (ELBO) on the data log-likelihood: a reconstruction term plus a KL term that regularises the latent posterior toward a prior:

$$\mathcal{L}_{ELBO} = \mathbb{E}_{q(z|x)}[\log p(x|z)] - D_{KL}(q(z|x) \parallel p(z))$$

With Gaussian posterior and standard normal prior, the KL term has a closed form, and sampling uses the reparameterisation trick so gradients flow through the latent:

$$D_{KL} = -\frac{1}{2} \sum_{j=1}^d (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2)$$

$$z = \mu + \sigma \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$

5.2 From pixels to features — the key decision

A from-scratch convolutional VAE (Notebook 03) was trained over six variants to reconstruct 128x128 pixels. All produced blurry reconstructions capturing average colour rather than fine style cues, and weak latents — because reconstructing raw pixels from ~1,000 images cannot teach the texture statistics that define style. The fix (Notebook 03b): feed the VAE *pretrained features* instead of pixels. A frozen backbone produces one feature vector per image; a small VAE reconstructs that vector, with an auxiliary class head shaping the latent. The total objective is:

$$\mathcal{L} = \mathcal{L}_{rec} + \beta \mathcal{L}_{KL} + \lambda \mathcal{L}_{CE}$$

Reconstruction (of the feature vector) + beta x KL + lambda x cross-entropy. The latent mu is exported as the embedding.

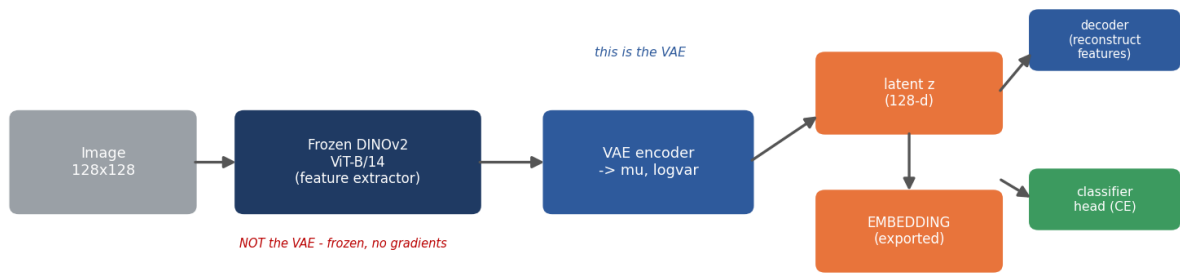


Figure 5. Feature-space VAE. DINOv2 is a frozen feature extractor (it is NOT the VAE); the VAE is the encoder-latent-decoder with the KL term, trained on those features. Its latent mu becomes the embedding used downstream.

The linear-probe score on the raw features is the diagnostic ceiling the latent can approach:

Feature source for the VAE	Linear-probe macro-F1	VAE-head test macro-F1
Raw pixels (from scratch)	n/a	~0.40
ResNet-18 features (512-d)	0.731	0.717
DINOv2 ViT-B/14 features (768-d)	0.848	0.818

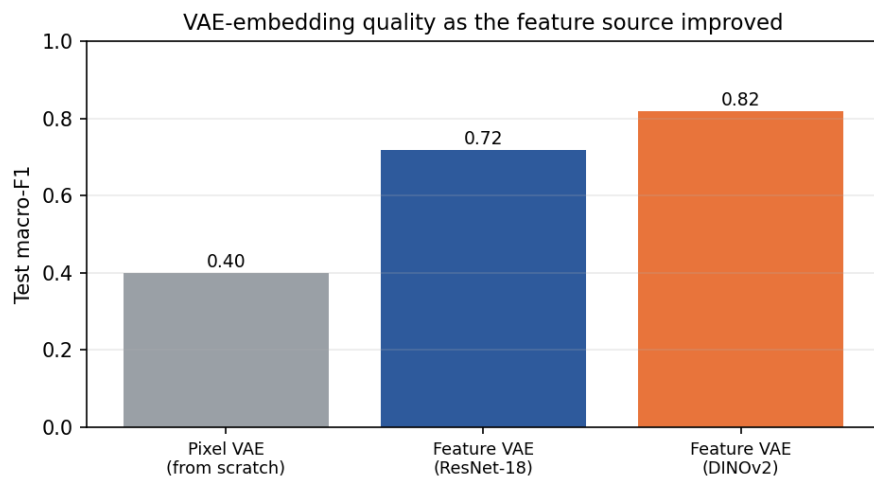


Figure 6. The VAE stayed fixed; only the feature source changed. Embedding quality more than doubled, proving the representation - not the VAE hyperparameters - was the bottleneck.

5.3 Does the VAE work? (reconstruction analysis)

Since this VAE reconstructs feature vectors, it is evaluated in feature space. Reconstruction is deliberately lossy (R-squared 0.21, mean cosine 0.47 - the 128-d latent compresses 768-d features), yet the class-relevant signal survives: nearest-neighbour retrieval gives 0.72 label-agreement from reconstructed features vs 0.74 from originals. The bottleneck discards reconstruction detail but keeps style - the right trade-off for a downstream embedding.

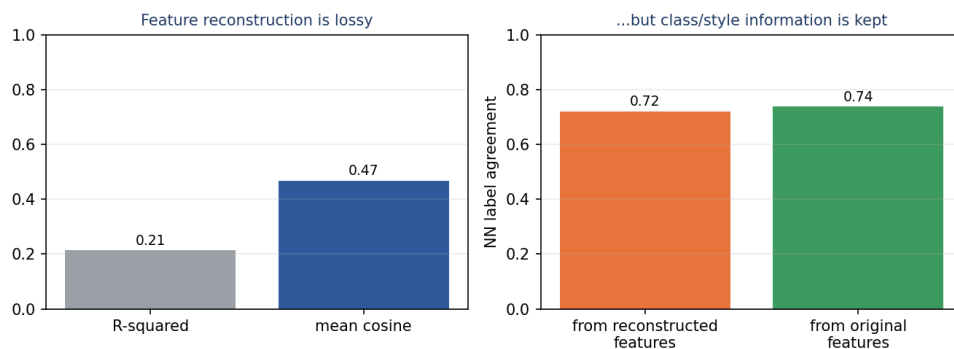


Figure 7. Left: feature reconstruction is lossy. Right: but the class/style information is almost fully preserved through the bottleneck (0.72 vs 0.74 label agreement).

6. Task 2 — Clustering and Latent Geometry (Notebook 04)

Why these algorithms (latent geometry). The KL term pushes the posterior toward an isotropic Gaussian prior. This makes distances roughly Euclidean (suited K-Means, which assumes spherical equal-variance clusters) and the space literally Gaussian-mixture-friendly (suited a GMM with full per-cluster covariance). Both were compared and scored against the true labels.

Clustering quality is judged by the silhouette coefficient (cluster cohesion vs separation) and, against ground truth, by Normalised Mutual Information and the Adjusted Rand Index:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

$a(i)$ = mean intra-cluster distance, $b(i)$ = mean nearest-other-cluster distance.

$$\text{NMI}(U, V) = \frac{2 I(U; V)}{H(U) + H(V)}$$

Mutual information $I(U;V)$ normalised by the cluster/label entropies; ARI similarly corrects the Rand index for chance.

Clustering was run on the *unsupervised* pixel-VAE latents (the honest, non-circular choice). Separation there is modest, matching the weak pixel representation; by contrast the class-regularised DINOv2 latents show clear per-style structure in t-SNE/UMAP, which is what makes Task 3 strong. Interpretation focused on the dominant style per cluster and the visual factors (palette, texture, geometry) that separate them.

7. Task 3 — Gradient Boosting vs the Neural Network (Notebook 05)

Gradient boosting builds an additive ensemble of trees, each fitting the negative gradient of the loss; multiclass outputs go through a softmax with log-loss:

$$F_M(x) = \sum_{m=1}^M \nu h_m(x), \quad h_m \approx -\nabla_F \mathcal{L}$$

$$\hat{p}_c = \frac{e^{F_c(x)}}{\sum_k e^{F_k(x)}}$$

Trained on the VAE embeddings and compared with the neural network on the *same* embeddings (isolating classifier from representation):

Model	Accuracy	Macro-F1	Bal. Acc.
XGBoost (best overall)	0.874	0.865	0.858
HistGradientBoosting	0.860	0.848	0.835
Neural net (VAE head)	0.832	0.818	0.813

Boosting beats the neural net by 3-5 macro-F1 points on identical inputs; XGBoost is the best model in the project (0.865). Feature relevance is measured by permutation importance — the drop in score when a latent dimension is randomly shuffled:

$$\text{Imp}_j = s - \frac{1}{R} \sum_{r=1}^R s_{\pi_j^{(r)}}$$

s = baseline score; the j -th feature's values are permuted R times and the averaged score drop is its importance. Only ~15 of 128 latent dims carry most of the signal.

7.1 How much is the representation worth?

Repeating the comparison on the unsupervised pixel-VAE embeddings quantifies the representation's contribution (and addresses honestly that the DINOv2 latents are label-aware):

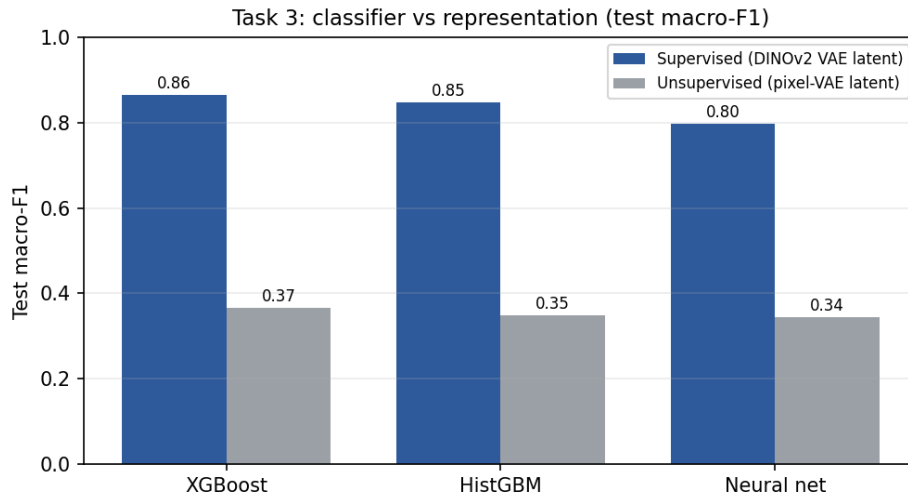


Figure 8. The supervised-vs-unsupervised gap (~0.50 macro-F1) dwarfs the boosting-vs-neural-net gap within either group. Once the representation is good, the classifier choice matters far less than where the features came from.

The unsupervised embeddings reach only ~0.35-0.37 macro-F1 (well above the 0.125 random baseline, but far below the feature VAE) — the empirical, numerically anchored justification for the whole pixel-to-feature-space decision.

8. Explainable AI Across the Pipeline

Explainability appears in all three tasks. For the classifier, Grad-CAM weights the final convolutional feature maps by the gradient of the predicted class, producing a spatial importance heatmap:

$$\alpha_k = \frac{1}{Z} \sum_{i,j} \frac{\partial y^c}{\partial A_{ij}^k}, \quad L^c = \text{ReLU}(\sum_k \alpha_k A^k)$$

α_k = gradient-based weight of feature map k ; the ReLU keeps only evidence that supports the predicted class c .

What Grad-CAM revealed. On correct predictions the model attends to figures and stylised poster elements, not background. The misclassifications are the insight: ArtDeco mistaken for Japonism occurs when the heat sits on an ornamental fan/peacock motif that genuinely resembles Japonism, and ArtDeco mistaken for Cubism occurs on hard angular drapery that reads as Cubist faceting. Errors come from real cross-style overlap, not spurious cues — validating the classifier and explaining the confusion structure.

- **Task 2:** t-SNE / PCA / UMAP latent projections coloured by class, plus a nearest-neighbour reconstruction proxy visualising what the feature-VAE latent preserves.
- **Task 3:** permutation importance over latent dimensions, and confusion matrices at every stage.

9. Conclusions

Stage	Best approach	Test macro-F1
Task 1 - classifier	ResNet-18 transfer learning	0.732
Task 1 bonus - SSL	SimCLR fine-tuning	0.468
Task 2 - VAE embeddings (NN head)	DINOv2 feature-space VAE	0.818
Task 3 - boosting on embeddings	XGBoost	0.865

Decisions that mattered most

- Choosing macro-F1 / balanced accuracy over accuracy, driven by the 5.6:1 imbalance from EDA.
- Switching from a from-scratch CNN to ImageNet transfer learning (+0.238 macro-F1).
- Rebuilding the VAE in feature space on frozen DINOv2 features — roughly doubling embedding quality and lifting every downstream task at once.
- Letting evidence pick the final classifier: gradient boosting on the embeddings beat the neural net.

Honest limitations

- Small dataset; per-class metrics for ArtDeco (9 test images) are noisy.
- The strongest embeddings are class-regularised (label-aware); this is stated transparently and the unsupervised-vs-supervised comparison in Task 3 quantifies exactly how much it helps.
- SimCLR is under-powered at this data scale and is a demonstration, not a competitive model.

Reproducibility

A git repository runs notebooks 01-05 in order (03b depends on 03; 04 on the VAE embeddings; 05 on 03b). Each stage saves checkpoints, per-epoch histories, classification reports, confusion matrices and figures under models/, outputs/, reports/figures/, with Google Drive backups. Large artifacts (checkpoints, embedding .npz) are excluded from git and regenerated by running the matching notebook.